

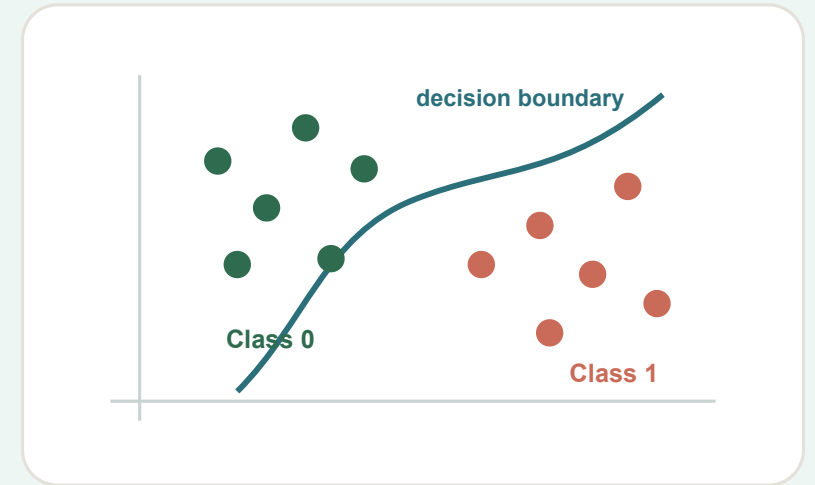
Classification Algorithms

Logistic regression, LDA, QDA, and KNN

How to choose between them

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Classification Methods

01

Classification recap

What classification is and a quick logistic regression review

02

LDA

Linear Discriminant Analysis

03

QDA

Quadratic Discriminant Analysis

04

KNN

K Nearest Neighbors

05

Compare the methods

When each classifier works well

FROM LAST WEEK

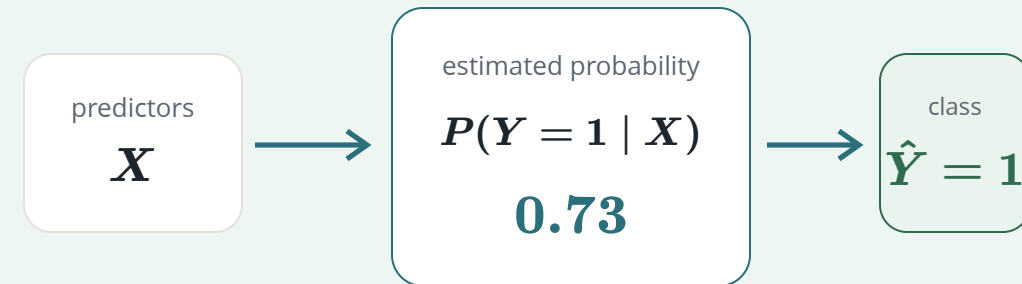
Logistic regression estimates a probability

Given predictors X , logistic regression estimates the probability that an observation belongs to Class 1:

$$P(Y = 1 \mid X)$$

We then compare that probability with a threshold, often 0.5, to assign a class.

Next question: can we classify observations using a different idea?



Using threshold 0.5:

$$0.73 > 0.5 \Rightarrow \hat{Y} = 1$$

The same observation can get different predictions

Logistic regression, LDA, QDA, and KNN use different assumptions and decision rules.

Logistic regression

PREDICTION

Default

Estimates $P(Y = 1 | X)$,
then applies a threshold

LDA

PREDICTION

Default

Uses class distributions
with a shared covariance
matrix

QDA

PREDICTION

No default

Allows each class to have
its own covariance matrix

KNN

PREDICTION

No default

Uses a vote among
nearby observations

The question is not only “**What does the model predict?**” but also “**Why does this method make that prediction?**”

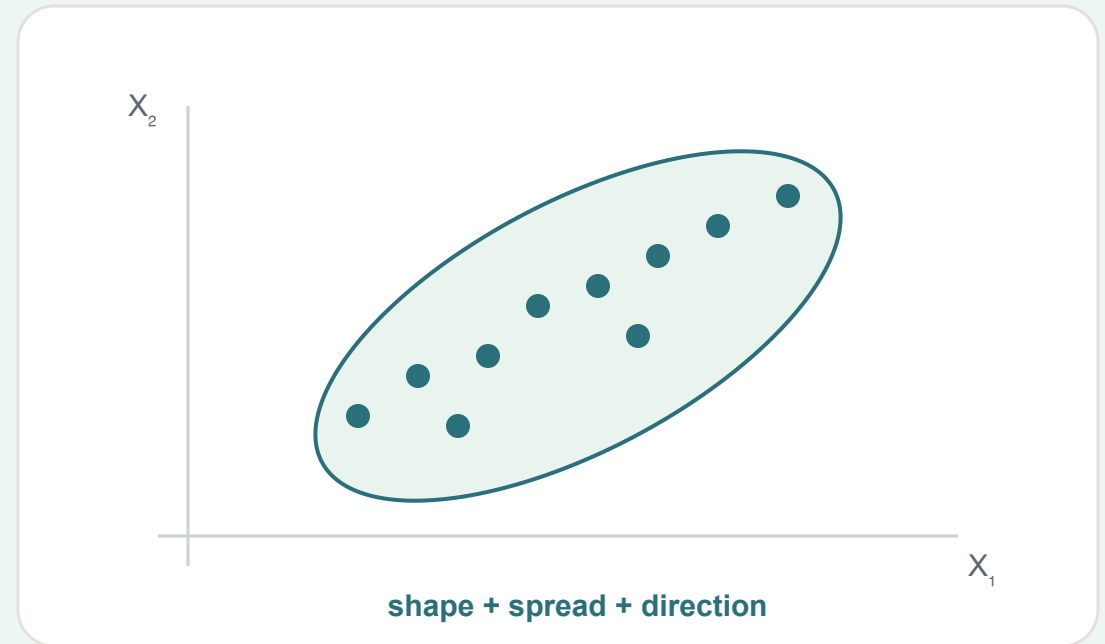
What is a covariance matrix?

For two predictors X_1 and X_2 :

$$\Sigma = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{pmatrix}$$

The diagonal tells us how much each variable varies.

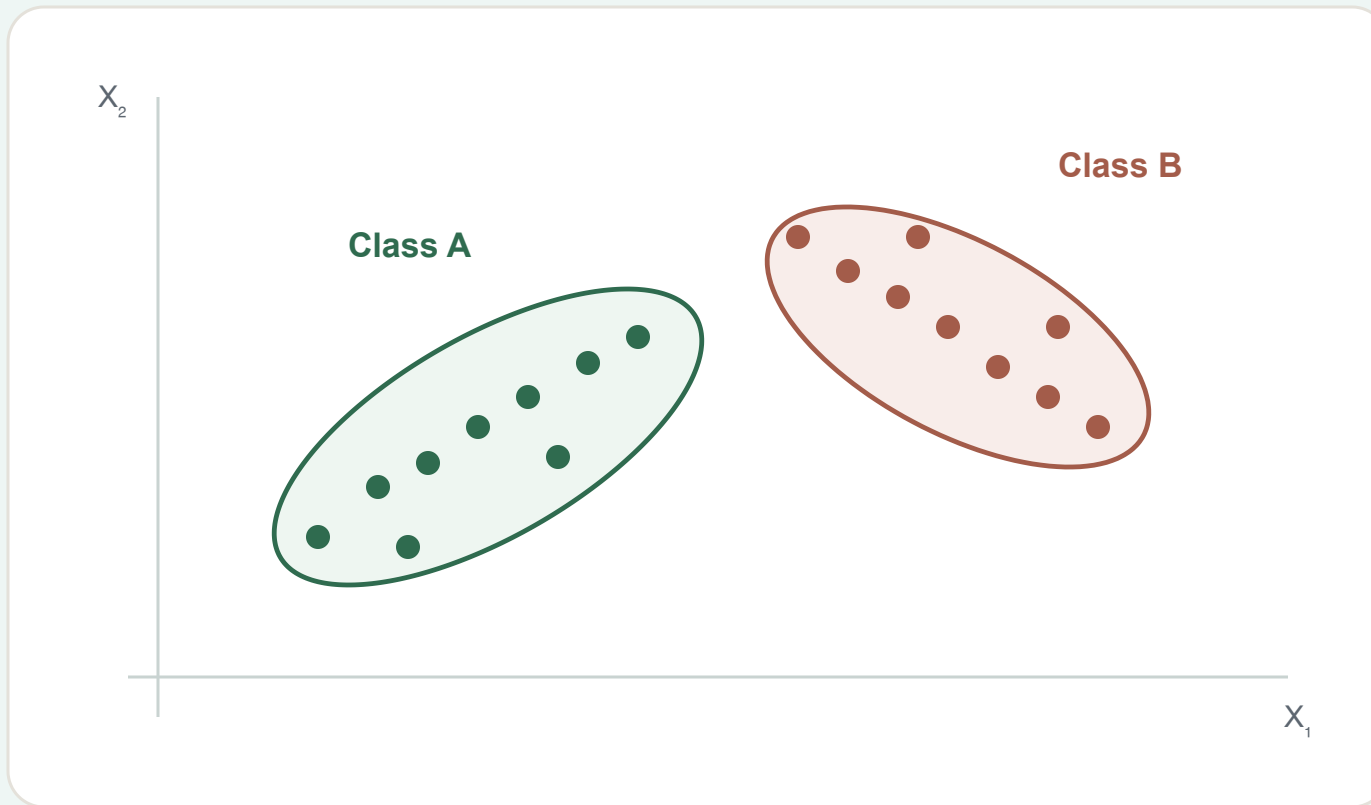
The off-diagonal tells us how the variables move together.



Think of the covariance matrix as describing the **shape of a cloud of data points**.

LDA assumes the classes share the same covariance matrix; QDA allows each class to have its own.

Two classes can have different shapes and directions



Class A

The cloud slopes downward from left to right.

$$\text{Cov}(X_1, X_2) < 0$$

Class B

The cloud slopes upward from left to right.

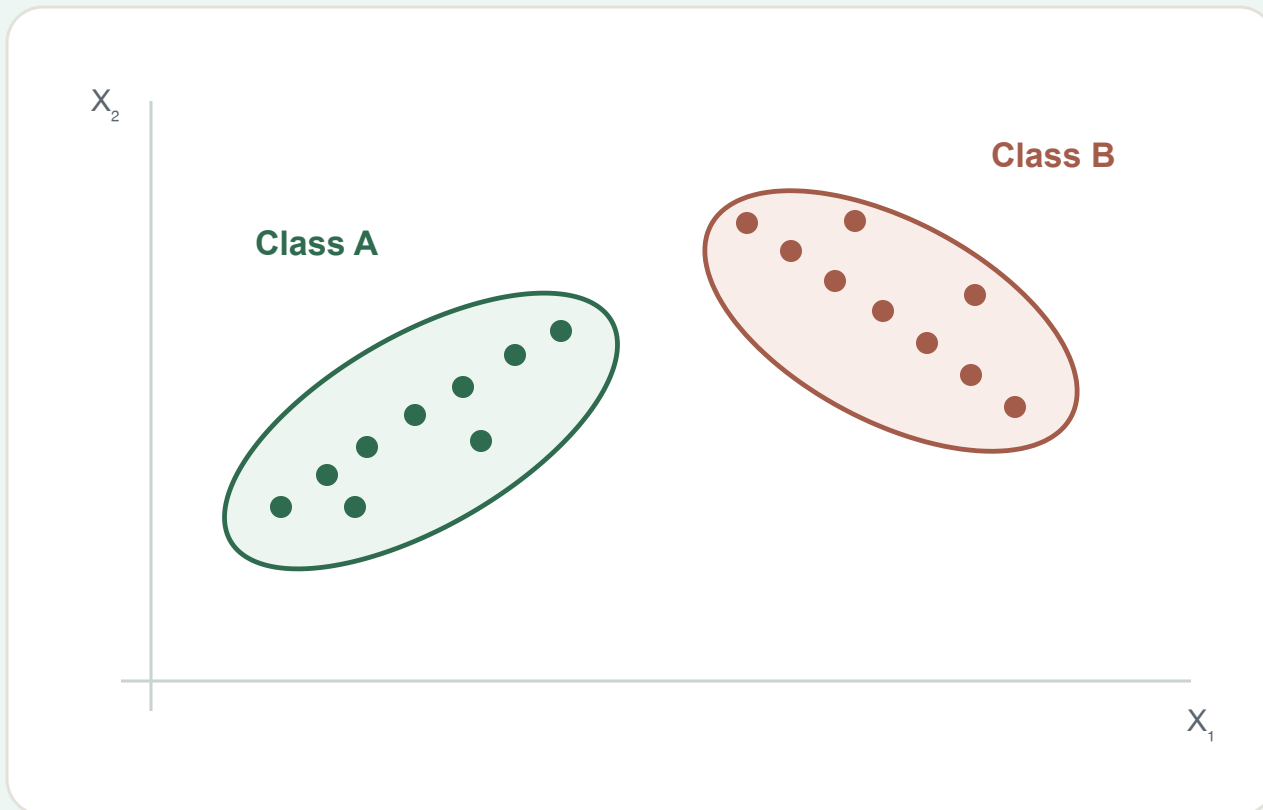
$$\text{Cov}(X_1, X_2) > 0$$

Different covariance matrices mean the classes can have different **spread, shape, and orientation**.

This is something QDA can model directly, while LDA assumes the classes share the same covariance structure.

Different covariance structures

Variance describes spread. Covariance describes how two variables move together. Together they determine the shape and orientation of a class.



KEY FORMULAS

Variance

Spread of one variable around its mean

$$\text{Var}(X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Covariance

How two variables move together

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Class A: $\text{Cov}(X_1, X_2) < 0$

As X_1 increases, X_2 tends to decrease.

Class B: $\text{Cov}(X_1, X_2) > 0$

As X_1 increases, X_2 also tends to increase.

Different covariance matrices: $\Sigma_A \neq \Sigma_B$

This is why **QDA** can be more flexible than **LDA**: LDA assumes a common covariance structure, while QDA allows each class to have its own shape and orientation.

FRAMING

Classification predicts a category, not a continuous number

Regression might predict **\$52,400**. Classification predicts a label such as **spam**, **default**, or **species**.

Binary classification

Spam or not spam

Two possible classes

Multiclass classification

Setosa, versicolor, virginica

More than two possible classes

The response variable Y is categorical: the goal is to decide which class an observation belongs to.

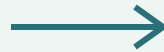
The Iris dataset: a multiclass example

Each observation is one iris flower. We measure parts of the flower and predict which of three species it belongs to.

PREDICTORS X

Flower measurements

- Sepal length
- Sepal width
- Petal length
- Petal width



RESPONSE Y

Species

Setosa

Versicolor

Virginica

Example: given a flower's measurements X , the classifier might predict $\hat{Y} = \text{versicolor}$.

Because there are three possible labels, this is a **multiclass classification** problem.

The species names are simply the category labels we are trying to predict.

RUNNING EXAMPLE

Credit-card default

Can we predict whether a customer will default using balance, income, and student status?

10,000

customers

One observation per customer

333

defaulted

Only 3.33% of customers

96.7%

accuracy

if we predict "No Default" for everyone

PREDICTORS X

Balance · Income · Student status

RESPONSE Y

Default / No Default

Important: high accuracy does not always mean a useful classifier when one class is rare.

PART 1

Logistic regression

Model the probability of a class directly.

The log-odds are a straight line

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

Here, $p = P(Y = 1 \mid X)$.

A one-unit increase in X changes the **log-odds** by β_1 .

Equivalently, the **odds** are multiplied by e^{β_1} .

DEFAULT DATA · BALANCE ONLY

$$\hat{\beta}_0 = -10.65 \quad \hat{\beta}_1 = 0.0055$$

Higher credit-card balance increases the estimated log-odds of default.

From balance to probability of default

$$\log \left(\frac{p}{1-p} \right) = -10.65 + 0.0055X$$

STEP 1

Choose a balance

$$X = 2000$$

Suppose the customer has a credit-card balance of \$2,000.

STEP 2

Compute log-odds

$$\begin{aligned} & -10.65 + 0.0055(2000) \\ & = 0.35 \end{aligned}$$

STEP 3

Convert to probability

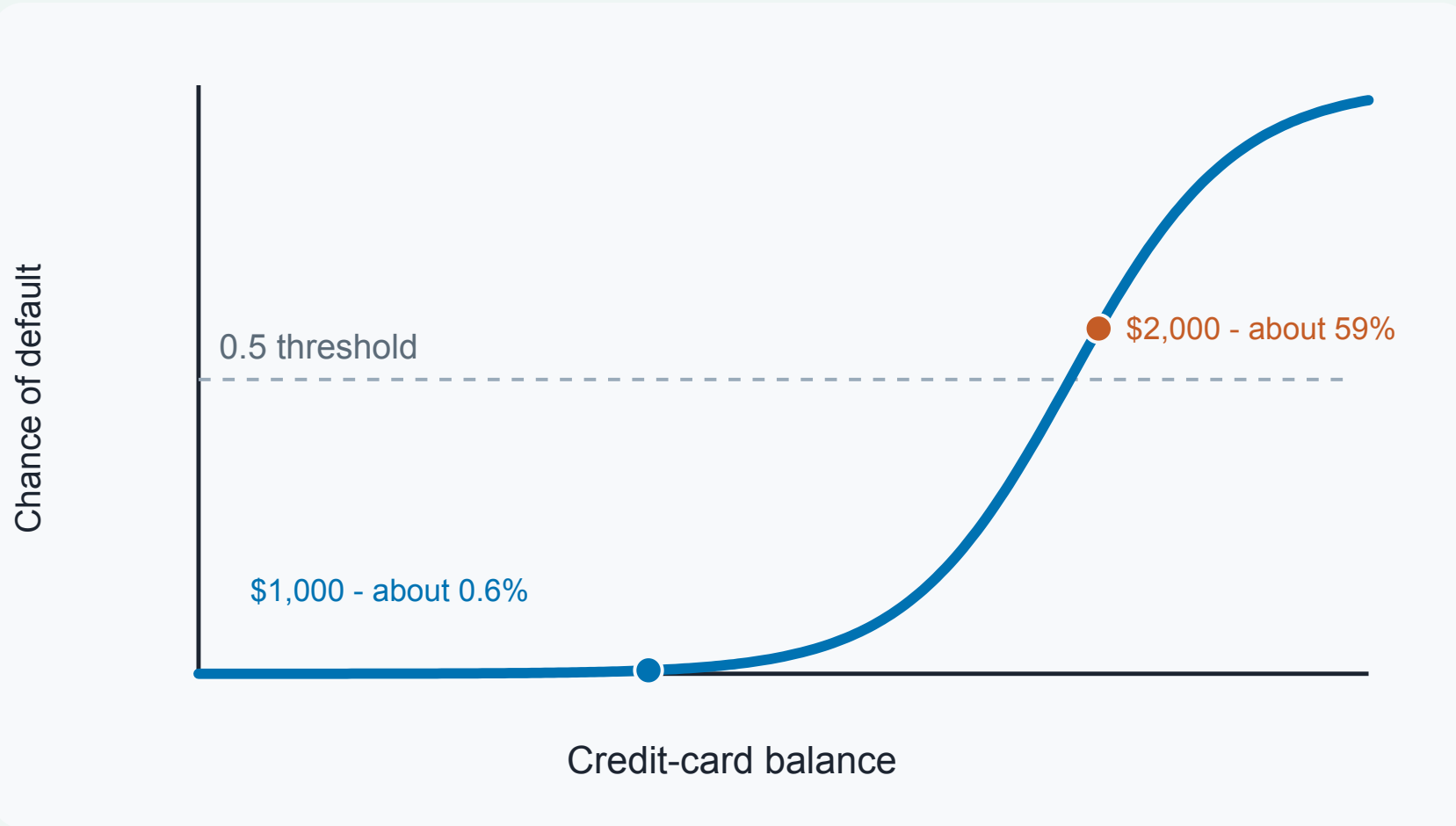
$$\begin{aligned} p &= \frac{e^{0.35}}{1+e^{0.35}} \\ p &\approx \mathbf{0.587} \end{aligned}$$

The model estimates about a **58.7% probability of default**.

With a threshold of 0.5, this customer would be classified as **Default**.

Balance → **log-odds** → **probability** → **predicted class**

Higher balance means a higher estimated probability of default



The linear model acts on the log-odds, but the resulting probability follows an S-shaped curve and always stays between 0 and 1.

The student coefficient changes sign

What looks true by itself can change after we control for another variable.

MODEL 1

Balance only

$$\hat{\beta}_{balance} = +0.0055$$

Higher balance is strongly associated with a higher probability of default.

\$1,000 balance → about **0.6%**

\$2,000 balance → about **59%**

MODEL 2

Student only

$$\hat{\beta}_{student} \approx +0.40$$

Looking only at student status, students appear more likely to default.

Students → about **4.3%**

Non-students → about **2.9%**

MODEL 3

Balance + student

$$\hat{\beta}_{student} \approx -0.65$$

After controlling for balance, the student effect reverses.

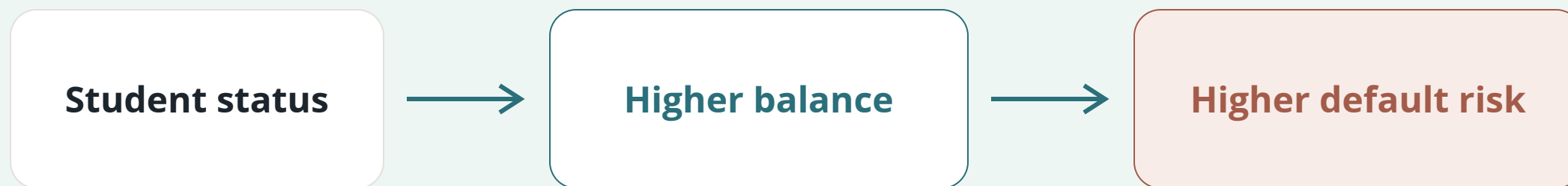
At the **same balance**, students are estimated to have lower default risk.

WHY DOES THE SIGN FLIP?

Students tend to have higher balances, and higher balance strongly predicts default. Without controlling for balance, some of the **balance effect gets attributed to student status**.

WHY THE FLIP?

Students tend to carry higher balances



If we use **student status alone**, some of the balance effect gets attributed to being a student.

Once balance is included in the model, we compare students and non-students with similar balances.

HOLD BALANCE AND INCOME FIXED

At a balance of **\$1,500** and income of **\$40,000**:

Student → about **5.8%** default probability · Non-student → about **10.5%**

This is a confounding effect: the apparent student effect changes when balance is controlled for.

The apparent student effect is largely explained by balance

The comparison changes when we hold credit-card balance fixed.

Student only

Student looks riskier

Student  4.3%

Not a student  2.9%

Coefficient about +0.4

Balance is left out, so the student label absorbs it.

Same balance, \$1,500

Student looks safer

Student  5.8%

Not a student  10.5%

Coefficient about -0.6

Once balance is included, being a student lowers risk.

Marginal comparison: students appear riskier. · **Conditional comparison:** at the same balance, students appear less risky.

Model the class directly, or model how each class looks

DISCRIMINATIVE

Logistic regression

Directly models the probability of a class given the predictors.

$$P(Y | X)$$

Question: **Given this customer's X , which class is most likely?**

GENERATIVE

LDA and QDA

Model how the predictors are distributed inside each class.

$$P(X | Y) \quad \text{and} \quad P(Y)$$

Then use Bayes' theorem to obtain $P(Y | X)$.

$$P(Y | X) \propto P(X | Y) P(Y)$$

Logistic regression learns the class probability directly; LDA and QDA first learn the shape of each class.

PART 2

Linear discriminant analysis

Model what the predictors look like inside each class.

$$P(X | Y) \rightarrow P(Y | X)$$

Which class does this observation look more like?

LDA models the distribution of the predictors inside each class.

NO DEFAULT

Lower average balance

This class has its own center:

mean of No Default

But LDA assumes the same spread as the other class.

DEFAULT

Higher average balance

This class has a different center:

mean of Default

But LDA assumes the same spread as the No Default class.

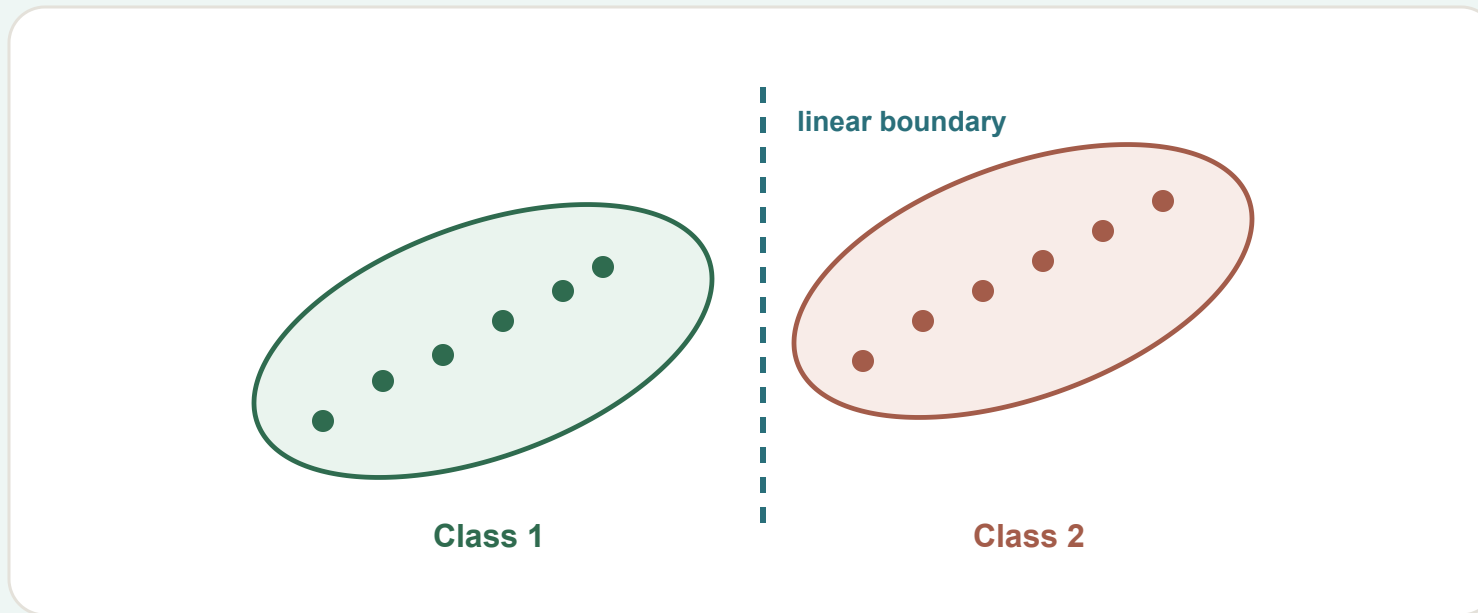
LDA ASSUMPTION

Different class centers, but the same variance in one dimension.

With several predictors, this becomes a shared covariance matrix.

LDA lets the classes have different centers, but assumes they have the same shape and spread.

Different centers. The same shape.



WHAT LDA ALLOWS

Different centers

Each class can live in a different location.

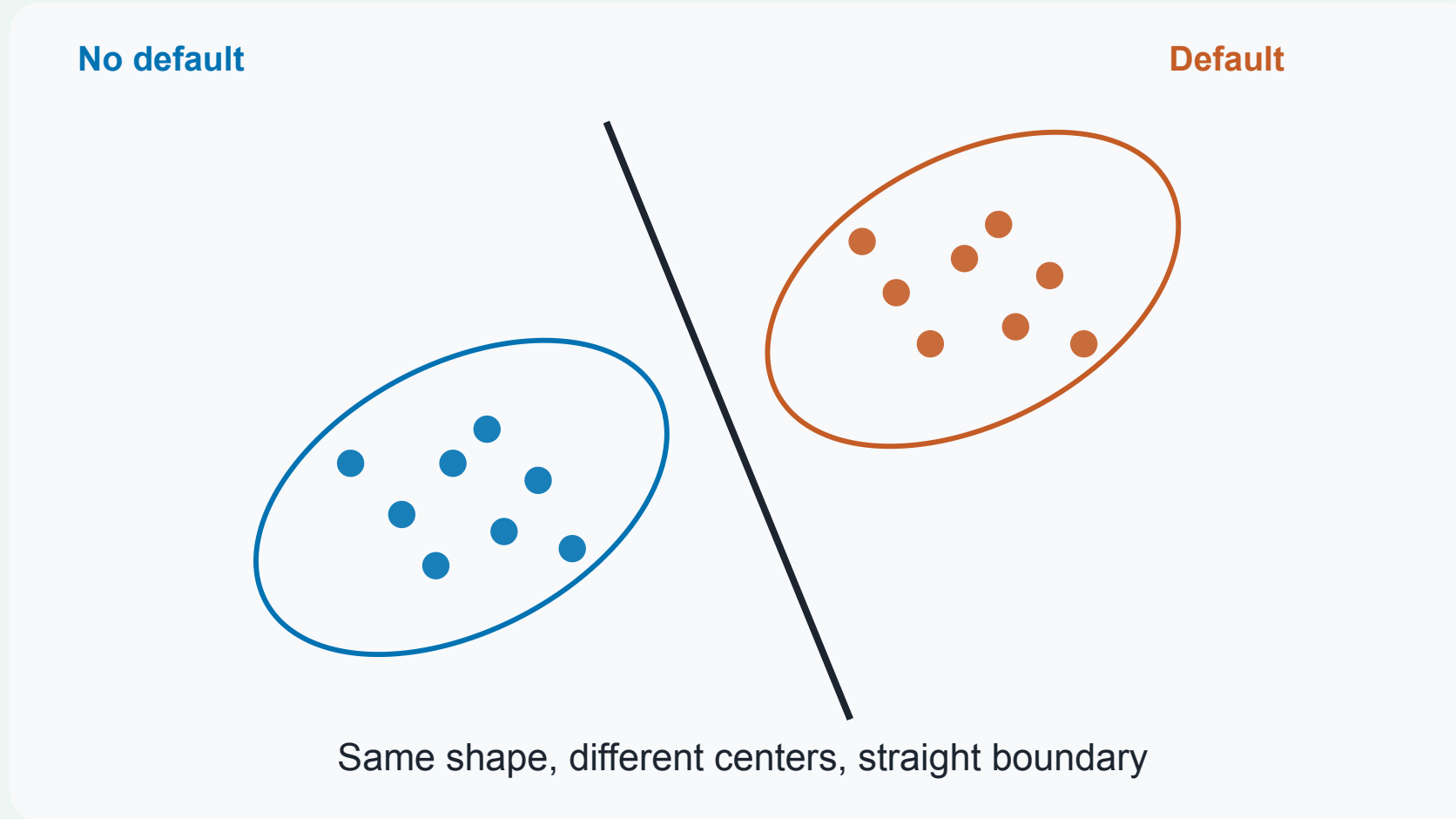
Same shape and spread

The classes share the same covariance structure.

Because the class shapes are shared, LDA produces a **linear decision boundary**.

If the classes truly have different shapes or spreads, a linear boundary may be too restrictive.

LDA separates the classes with a linear boundary



Shared class shape + different centers → a straight decision boundary.

PART 3

Quadratic discriminant analysis

Each class gets its own shape and spread.

Different covariance structures → curved decision boundaries

ONE KEY DIFFERENCE

LDA shares the shape; QDA does not

The difference is how much flexibility we allow each class to have.

LDA

One shared covariance

Every class is assumed to have the same overall shape, spread, and orientation.

→ Linear decision boundary

→ Fewer parameters to estimate

Often more stable when the sample size is limited.

QDA

One covariance per class

Each class can have its own shape, spread, and orientation.

→ Curved decision boundary

→ More parameters to estimate

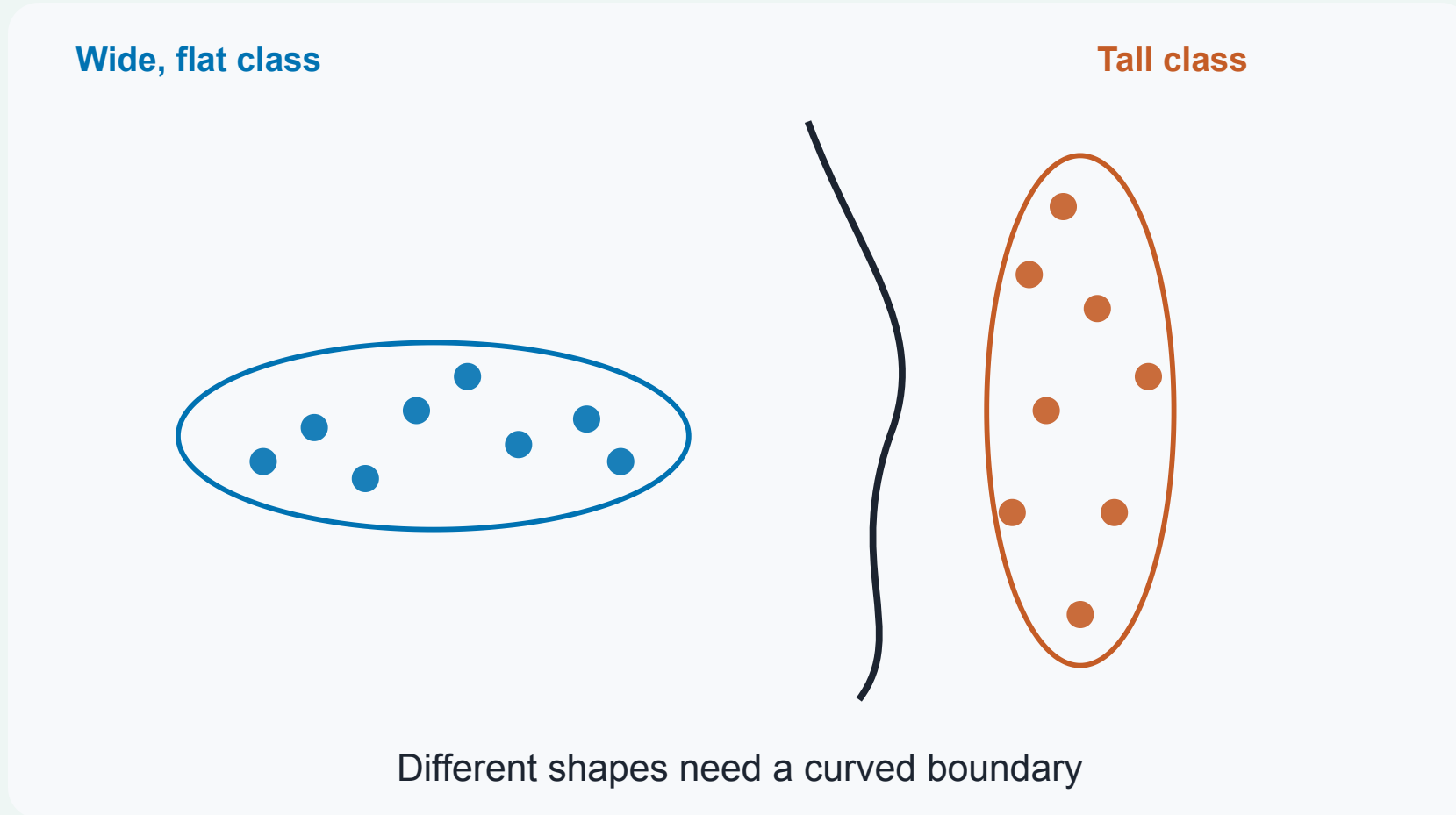
More flexible, but usually needs more data.

LDA = simpler and more constrained. · **QDA = more flexible but more data-hungry.**

This is a bias–variance tradeoff: extra flexibility can help when class shapes truly differ, but it can also increase estimation noise.

QDA can use a curved decision boundary

Different class shapes and spreads create a more flexible boundary.



Different covariance structures → **curved decision boundary**

THE TRADEOFF

More flexibility can help. It can also fit noise.

LDA

Simpler, more stable

Good when the classes have similar covariance structure or the sample size is limited.

QDA

More flexible, more variable

Useful when class shapes truly differ and there is enough data to estimate them reliably.

Bias-variance tradeoff: LDA has more bias but lower variance; QDA has less bias but higher variance.

Two small datasets: LDA-friendly vs QDA-friendly

LDA-FRIENDLY EXAMPLE

The two classes have similar spread; they are mainly shifted apart.

Class	Income	Balance
No Default	45	800
No Default	52	900
No Default	48	1100
No Default	55	1200
Default	42	1800
Default	50	2000
Default	47	2100
Default	54	2300

Main difference: **location**, not shape.

QDA-FRIENDLY EXAMPLE

The Default class is much more spread out and has a different pattern.

Class	Income	Balance
No Default	45	800
No Default	52	900
No Default	48	1100
No Default	55	1200
Default	30	1400
Default	40	1800
Default	55	2400
Default	70	3200

Main difference: **location + spread + shape**.

Similar covariance structure → **LDA** · Clearly different covariance structure → **QDA**

PART 4

K-nearest neighbors

No fitted equation. Let the nearest neighbors vote.

Nearby observations → majority class → prediction

Classify by a vote

1

Choose K

Decide how many nearby training observations will vote.

Example: $K = 5$

2

Find the nearest

Measure distance in predictor space and identify the K closest observations.

Often: Euclidean distance

3

Let them vote

Predict the class receiving the most votes among the neighbors.

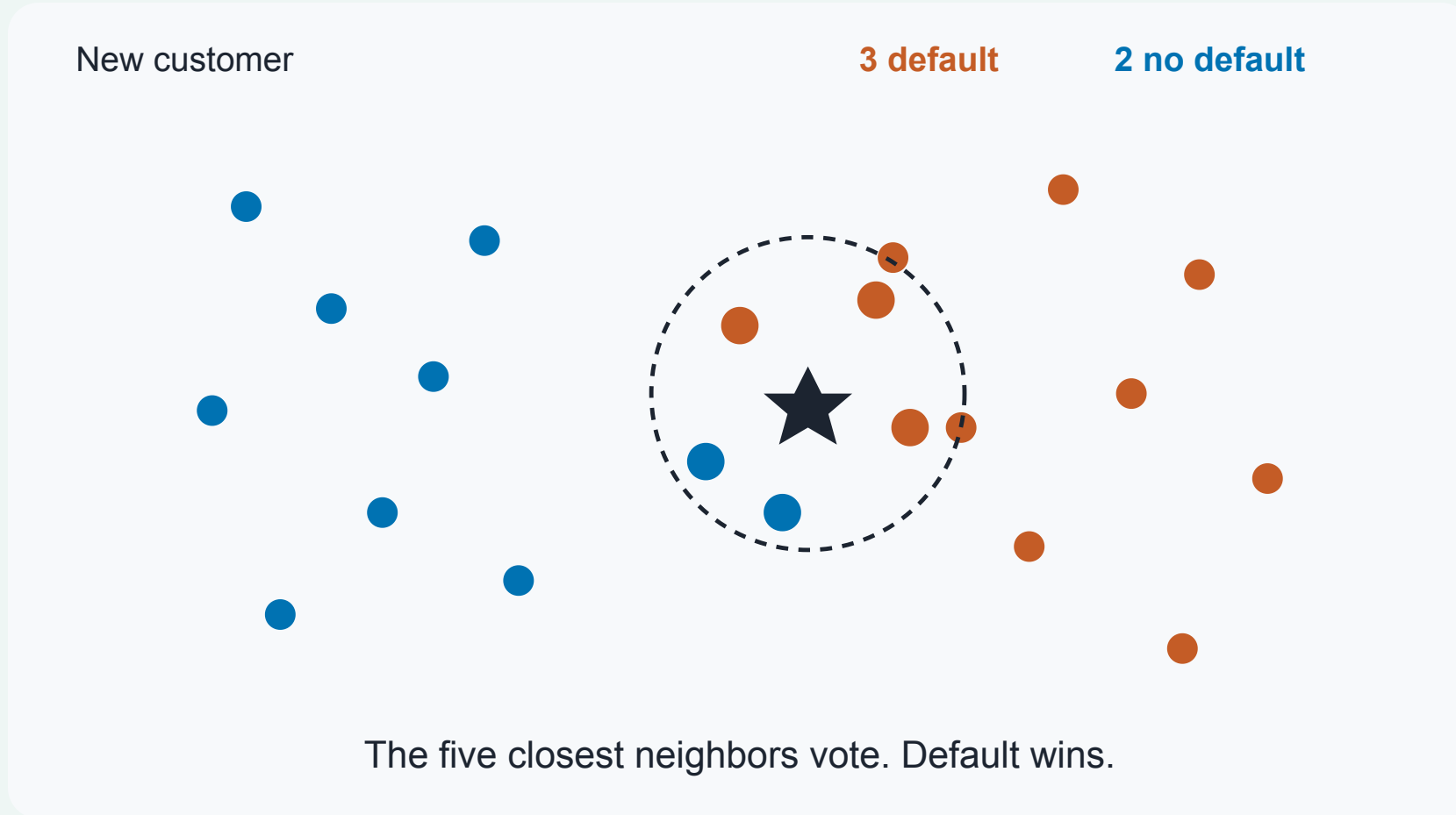
4 of 5 Default → predict Default

The local class proportion can also be used as an estimated probability.

<https://www.youtube.com/embed/gS9E7E0qOlc?enablejsapi=1>

Find the neighbors closest to the new customer

With $K = 5$, the five nearest training observations get the vote.



The majority class among the nearest neighbors becomes the prediction.

K = 5

Default wins the vote, 3 to 2

Each of the five nearest neighbors contributes one vote.



Default: **3 votes** · No Default: **2 votes**

Predict: Default

Local class proportion: 3 out of 5 = 60%

K controls how flexible KNN is

Small K reacts to very local patterns; large K smooths them away.

K = 1

Very flexible

Each observation is classified using only its single nearest neighbor.

- Jagged decision boundary
- Low bias
- High variance

Can easily fit noise.

Moderate K

Balance the tradeoff

Use several neighbors so individual noisy points have less influence.

- Smoother boundary
- Lower variance
- Some added bias

Choose K using validation or cross-validation.

K = n

Almost no flexibility

Every training observation votes for every new point.

- Very smooth boundary
- High bias
- Low variance

Predicts the majority class everywhere.

Small K → **more flexible** · Large K → **smoother and less flexible**

The best K is a data-driven choice, not a fixed rule.

Small K creates a flexible boundary; large K smooths it out

Increasing K makes the prediction depend on a larger neighborhood.

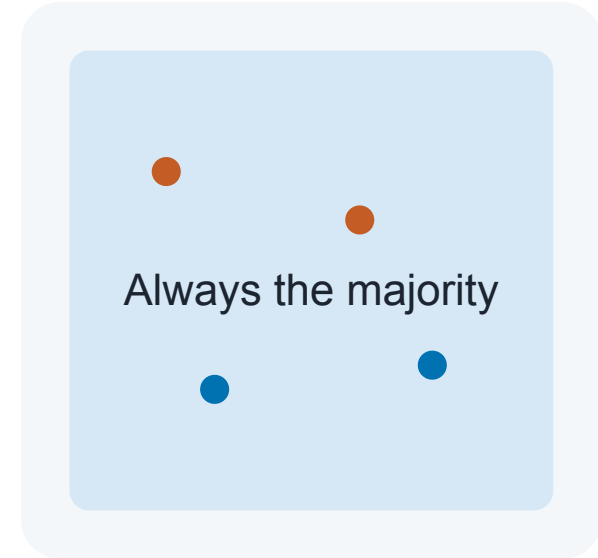
$K = 1$



Middle K



$K = n$



Small K → **local and wiggly** · Large K → **global and smooth**

AN IMPORTANT NAME

KNN is non-parametric

PARAMETRIC METHODS

Logistic regression, LDA, QDA

Assume a particular mathematical form and estimate a fixed set of model parameters from the training data.

Training data → estimate model → use fitted model to predict

NON-PARAMETRIC METHOD

K-nearest neighbors

Does not assume one fixed functional form for the decision boundary.

Keep training data → find neighbors → predict locally

Non-parametric does not mean “no choices.” KNN still requires choices such as K, the distance metric, and how predictors are scaled.

KNN works best when there is enough nearby data and the number of predictors is not too large.

PART 5

Which method?

Match the method to the data, the boundary, and the amount of flexibility you need.

Compare assumptions → validate performance → choose the model

Where each method fits

Logistic

Direct probability

- Interpretable probabilities
- Linear log-odds
- Simple global model

LDA

Shared class shape

- Linear boundary
- Same covariance
- Good with less data

QDA

Different class shapes

- Curved boundary
- Covariance per class
- Needs more data

KNN

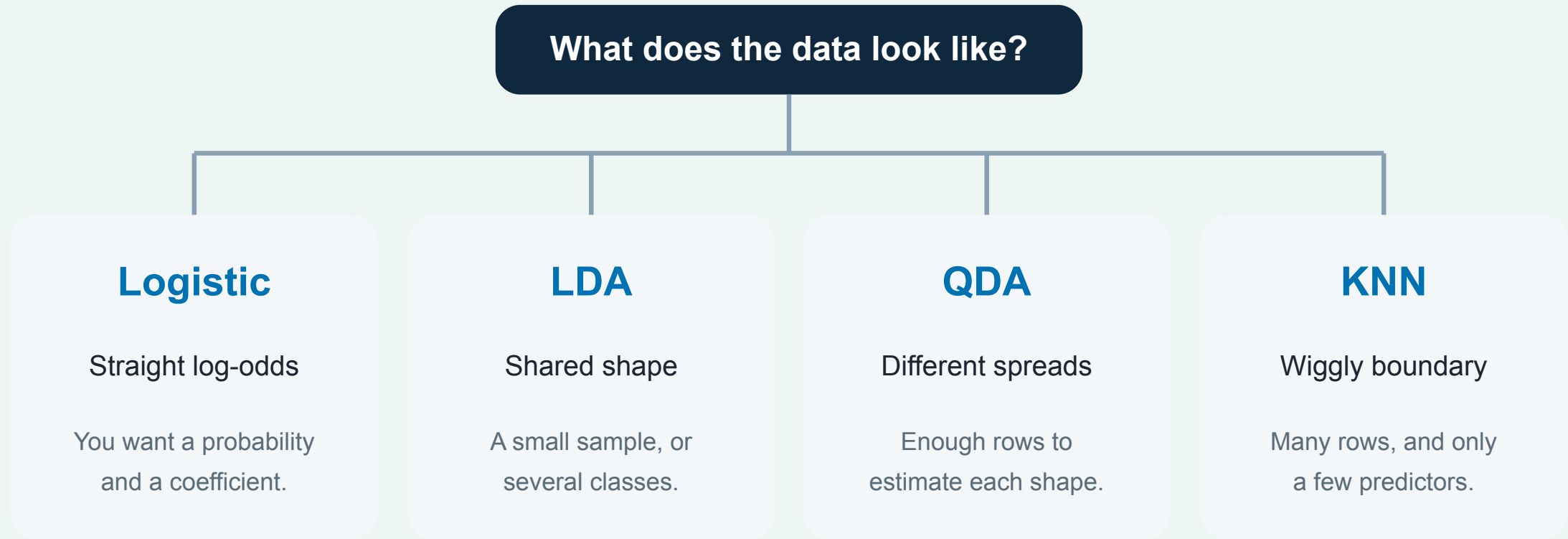
Local neighbors

- Flexible boundary
- Few predictors
- Scale matters

Start with the assumptions — then compare models using validation.

Start with the structure of the data

Ask what kind of boundary and class structure seem plausible.



Use the map as a starting point — then confirm the choice with validation.

Which method would you try first?

A

60 patients

- 8 lab measurements
- 3 diagnoses
- Roughly bell-shaped

B

50,000 rows

- 2 predictors
- Curved boundary
- Local patterns

C

Text classification

- Thousands of features
- Mostly zeros
- High-dimensional

Choose: **Logistic • LDA • QDA • KNN**

Be ready to explain your choice.

A reasonable first choice

A

LDA

- Small sample
- 3 classes
- Roughly Gaussian

B

KNN

- Lots of data
- Only 2 predictors
- Irregular boundary

C

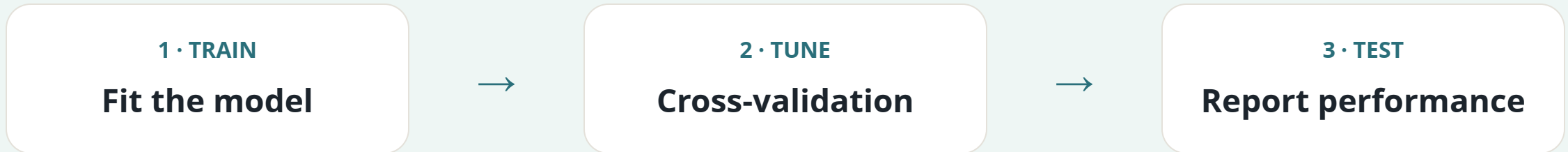
Logistic

- Many predictors
- Sparse features
- KNN struggles in high dimensions

These are starting choices — **validation decides.**

BEFORE YOU TRUST A SCORE

Judge the model on data it has not seen



Training accuracy can be misleading: “Always No Default” gets 96.7%, and KNN with $K = 1$ can memorize the training data.

Five takeaways

- 1** Logistic regression models class probability directly.
- 2** LDA assumes shared class shape and gives a linear boundary.
- 3** QDA allows different class shapes and can curve.
- 4** KNN votes locally; K controls flexibility.
- 5** Judge models on unseen data, not training accuracy.

READINGS

Chapter 4, and a first look at classifiers

ISLP, Chapter 4. Müller and Guido, Chapter 2.